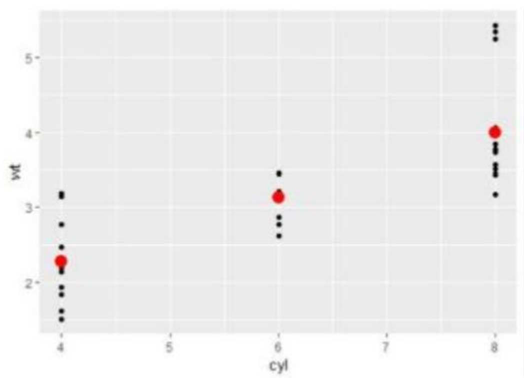
COP2073C Module 11 Programming Assignment

For this assignment you will need to install and load the tidyverse and modelr packages.

In this exercise, we will use tidyverse, modelr, and base R functions to practice analyzing a linear relationship with a categorical predictor.

Instructions

1. Create a tibble using the cyl (cylinders) and wt (weight) columns of the mtcars dataset. Note that cyl is a **categorical** variable.
2. Use the lm function to fit a linear model to predict wt based on cyl (wt ~ cyl).
3. Create a data grid from the tibble's cyl column using data_grid, and add predictions using this grid **(you must use pipes to chain these steps).**
4. Plot the original data and the predicted values as shown here, with the predicted values in red and size 4 for emphasis.



5. Provide your analysis of the results as commented lines at the end of your script. Use the analysis provided in the practice exercise as an example of the length and content for your analysis.

Non-Functional Requirements:

- Include a 4-line ID header at the beginning of your script.
- Include vertical spacing (a blank line) between logical blocks for readability.
- Comment your code thoughtfully (avoid excessive commenting).
- Ensure each line of code does not exceed 80 columns